

From Individual to Society: A Survey on Social Simulation Driven by Large Language Model-based Agents

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Traditional sociological research often relies on human participation, which, though effective, is expensive, challenging to scale, and with ethical concerns. Recent advancements in large language models (LLMs) highlight their potential to simulate human behavior, enabling the replication of individual responses and facilitating studies on many interdisciplinary studies. In this paper, we conduct a comprehensive survey of this field, illustrating the recent progress in simulation driven by LLM-empowered agents. We categorize the simulations into three types: (1) *Individual Simulation*, which mimics specific

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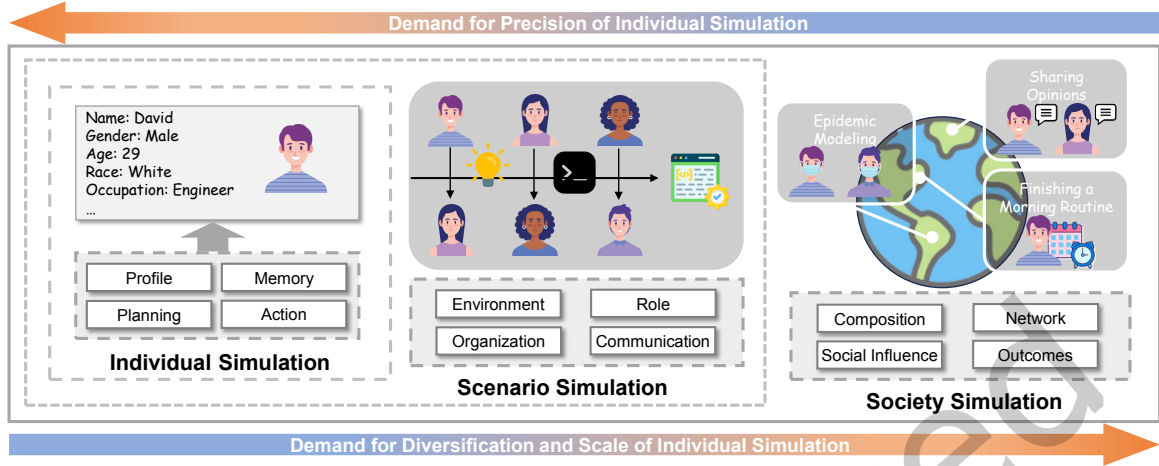


Fig. 1. Illustration of simulations empowered by LLM-driven agents. We categorize the simulations into individual simulation, scenario simulation and society simulation. From left to right, the diversity and scale of individual modeling generally increase. Conversely, from right to left, the granularity of individual modeling becomes more refined.

individuals or demographic groups; (2) *Scenario Simulation*, where multiple agents collaborate to achieve goals within specific contexts; and (3) *Society Simulation*, which models interactions within agent societies to reflect the complexity and variety of real-world dynamics. These simulations follow a progression, ranging from detailed individual modeling to large-scale societal phenomena. We provide a detailed discussion of each simulation type, including the architecture or key components of the simulation, the classification of objectives or scenarios, and the evaluation method. Afterward, we summarize commonly used datasets, benchmarks and tools. Finally, we discuss the risks and challenges across these three types of simulation. A repository for the related sources is at <https://github.com/FudanDISC/SocialAgent>.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Modeling and simulation**; • **Applied computing** → **Law, social and behavioral sciences**.

Additional Key Words and Phrases: social simulation, large language models, agent

1 Introduction

Social science investigates human behavior and social structures to understand how societies function. Traditional sociological research heavily relies on human participation to conduct experiments and gather data. Questionnaires [64, 100] and psychological experiments [143] are commonly used to test theoretical hypotheses, understand social phenomena, and predict collective outcomes. While these methods can provide highly authentic data, they are expensive, challenging to scale, and involve certain ethical risks.

Recently, large language models (LLMs) have demonstrated impressive capabilities in human-level reasoning and planning [101, 211, 225, 237, 257]. They can perceive the environment, make decisions, and take corresponding actions, showcasing their potential as autonomous agents that can serve as human substitutes. In appropriate settings, LLM-driven agents can accurately simulate responses from corresponding individuals by leveraging their role-playing abilities [29, 183], a property known as algorithmic fidelity [6, 24]. This characteristic makes LLM-driven agents highly valuable in simulating human behavior. By reproducing individual response patterns in specific scenarios, LLM-driven agents help researchers to better understand, validate, and predict human reactions.

Just as individuals do not exist independently within society, in addition to separate individual agents, interactions between multiple agents have also been widely studied to solve specific problems or simulate complex dynamics in the real world [57, 70]. On one hand, LLMs can be specialized as agents with detailed knowledge and

skills, leveraging collective intelligence to solve complex problems, such as software development [82, 162], automatic diagnosis [51, 114] and judicial decision-making [78]. In this case, multiple autonomous agents collaborate on planning, discussion, and decision-making, reflecting the cooperative nature of human groups when solving problems. On the other hand, simple interactions between multiple agents can lead to the emergence of complex collective behaviors or patterns [40, 79, 180], thereby replicating complex social dynamics in the real world, such as opinion dynamics [42, 137, 148] and macroeconomics phenomena [116]. Such simulations provide valuable tools for understanding, analyzing, and predicting complex phenomena that may be difficult or impractical to observe directly in real life, offering strong support for decision-making in areas such as policy-making and social management.

This research field is rapidly expanding, with papers focusing on various aspects. Considering the purpose of simulation and the varying demands for diversity, scale, and accuracy in individual modeling, we categorize the existing work into three types, as illustrated in Figure 1:

- (1) **Individual Simulation:** leveraging LLM-based agents to mimic specific individuals or groups of people sharing common demographic characteristics [29, 183, 224]. This line of research focuses on the replication of features of a single person, e.g., personality, and has not involved multi-agent interactions.
- (2) **Scenario Simulation:** organizing a group of agents in a concentrated scenario, driven by specific goals or tasks, such as software development [82, 162], question answering [49] and paper reviewing [45]. Such simulations are usually focused on small-scale agents within specific scenarios, emphasizing the collective wisdom of agents with specialized expertise.
- (3) **Society Simulation:** simulating more complex and diverse behaviors in the agent society to explore social dynamics in real-world applications. Such simulations could test social science theories within a small scope [43] or populate virtual spaces and communities with large-scale realistic social phenomena [157, 256]. The composition of individuals in such simulations is more complex and diverse.

These three types of simulations exhibit a progressive relationship. Individual simulation models a specific person or a type of person, serving as the foundation for scenario simulation and society simulation. Theoretically, society simulation can encompass a chaotic world composed of countless sub-scenarios.

Although this field has seen rapid growth, with some surveys summarizing agent architectures [57, 211, 237] or certain aspects of single-agent ability or multi-agent systems [29, 70, 132], there is an absence of a systematic review to summarize the work from the individual to society, providing a comprehensive blueprint for this field. This motivates us to present this survey, aiming to contribute to the research and development of simulations driven by LLM-based agents, as well as a wider range of interdisciplinary studies. To comprehensively describe our landscape, we organize our survey as follows. After a brief introduction to the background in § 2, we begin in § 3 by detailing how to conduct individual simulation through discussions of (1) the architecture of a single agent, (2) construction of individual simulation, (3) the classification of objectives, and (4) the evaluation of individual simulation. Next, in § 4, we summarize scenario simulation, including (1) the elements that constitute a scenario simulation system, (2) the classification of scenarios, and (3) the evaluation of scenario simulation, exploring how multiple agents collaborate to achieve objectives within a single scenario. Following this, in § 5, we introduce society simulation, examining how multi-agent systems can construct complex social dynamics through (1) the social construction elements of society simulation, (2) the classification of society simulation scenarios, and (3) the evaluation of society simulation. In § 6, we summarize existing datasets, benchmarks and tools. Then, we discuss potential risks and challenges in § 7 and § 8, and present the conclusion in § 9.

2 Background

This section provides the conceptual foundations for the taxonomy and discussions presented in the remainder of this survey. Rather than offering an exhaustive review, we focus on clarifying the core building blocks that underpin different forms of simulation.

2.1 Large Language Model-based Agents

An agent is an autonomous entity with its own attributes, decision rules, and goals, capable of perceiving its environment and interacting with others within a complex system [5, 81]. Traditional agent-based modeling (ABM) typically uses rule-based or heuristic entities that interact with each other and their environments. Such models have been widely used to study how micro-level behavioral rules give rise to emergent system-level patterns across domains including social science, economics, and transportation [15, 50, 169]. In these settings, agents often act as simplified proxies for humans, capturing limited aspects of individual beliefs, desires, and intentions.

While effective in modeling emergence, rule-based agents face notable limitations: restricted decision-making and planning, assumptions of perfect rationality, limited representation of diversity, and poor generalizability beyond handcrafted domains [57, 81, 237]. These constraints hinder adaptability in dynamic, open-ended environments.

Recent advances in large language models (LLMs) extend traditional agent-based modeling by enriching the cognitive and behavioral capacities of agents. Trained on large-scale data and aligned with human instructions, LLMs exhibit strong reasoning, in-context learning, and instruction-following abilities. These capabilities enable agents to move beyond reactive, rule-based behaviors and instead exhibit goal-directed actions supported by stepwise reasoning and adaptive decision-making [29, 70, 138, 211]. In this survey, LLM-based agents form the basic modeling unit in this survey. When considered in isolation, they support individual simulation. When placed in interactive settings, they become the constituent individuals in scenario and society simulations.

2.2 Multi-agent Systems

Modeling complex social phenomena often requires interacting populations rather than isolated agents. Multi-agent systems (MAS) consist of multiple autonomous agents interacting with one another and a shared environment [15, 50]. Traditional MAS typically employs rule-based or heuristic agents, whose repeated interactions can give rise to emergent patterns. This paradigm has been widely used to study interaction-driven social dynamics, such as coordination, competition, negotiation, and collective decision-making.

Despite their effectiveness under simplified assumptions, traditional MAS face fundamental limitations. Handcrafted rules and fixed communication protocols restrict agents' adaptability, limit population heterogeneity, and constrain knowledge-intensive interactions. As a result, such systems often struggle to support realistic scenario- or society-level simulations that require coherent responses to shared contexts and sustained multi-agent interactions. Integrating large language models as agent reasoning engines substantially extends the expressive power of MAS. LLM-driven agents enable language-mediated coordination, flexible role adaptation, and collective sense-making, supporting both scenario-level simulations under shared constraints and society-level simulations in which large populations give rise to social norms, institutions, and macro-level dynamics.

3 Individual Simulation

Individual simulation focuses on designing a modular architecture that integrates individualized data for the construction of agents and simulating the specific objective with high fidelity. In this section, we first outline the basic architecture of the agent in the individual simulation with four key components in §3.1. Then, two construction methods are discussed in §3.2 to implement the integration of individualized data into objectives introduced in §3.3. The evaluation methods are examined from different perspectives in §3.4. We further present critical discussions in §3.5. The overall framework is presented in Figure 2 and representative works are summarized in Table 1.

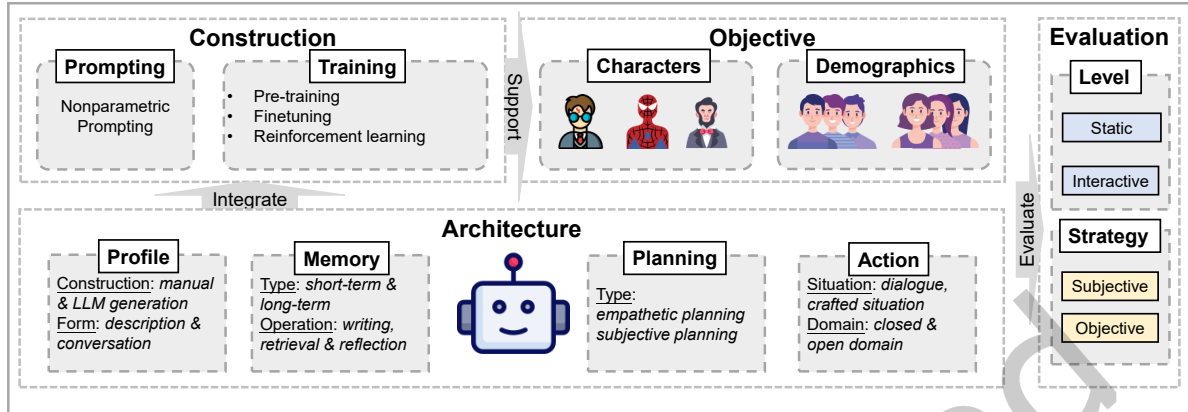


Fig. 2. Illustration of individual simulation blueprint. An individual agent is typically composed of an architecture with modules involving profile, memory, planning, and action through construction method, prompting or training, to simulate specific objectives like characters or demographics. Individual simulation can be evaluated statically and interactively with different dimensions.

Table 1. A list of representative works on individual simulation. In the Architecture column, we categorize each work according to its components, including profile, memory, planning, and action. In the Construction column, we further distinguish between parametric and non-parametric approaches.

Objectives	Paper	Architecture				Construction
		Profile	Memory	Planning	Action	
Characters	CharacterLLM [183]	Description	Short-term	Subjective	Open	Parametric
	Neeko [261], MMRole [44]	Description	Short-term	-	Open	Parametric
	RoleEval [185]	Description	Short-term	-	Closed	Parametric
	Yuan et al. [265]	Description	Short-term	-	Open/Closed	Nonparametric
	LifeChoice [247]	Description	Short/Long-term	-	Closed	Nonparametric
	Schwitzgebel et al. [182], ChatHaruhi [108]	Dialogue	Short-term	-	Open	Parametric
	Yu et al. [263]	Dialogue	Short-term	-	Open	Parametric
	CharacterEval [203]	Dialogue	Short-term	-	Open	Nonparametric
	Agrawal et al. [2], CharacterGLM [289]	Dialogue/Description	Short-term	-	Open	Parametric
	Brahman et al. [18], RoleLLM [224]	Dialogue/Description	Short-term	-	Open/Closed	Parametric
	Capturing Minds [171]	Dialogue/Description	Short/Long-term	Subjective	Open/Closed	Parametric
	LiveChat [59]	Dialogue/Description	Short/Long-term	-	Open/Closed	Parametric
Demographics	Rational Sensibility [193]	-	Short-term	Empathetic	Closed	Parametric
	InCharacter [217]	-	Short-term	-	Open/Closed	-
	Xie et al. [241]	Description	Short-term	Subjective	Closed	Nonparametric
	The wall street neophyte [242]	Description	Short-term	Empathetic	Closed	Nonparametric
	Out of One, Many [6], Toxicity in ChatGPT [46], Marked Personas [36], PersonaHub [61]	Description	Short-term	-	Open	Nonparametric
	Random Silicon Sampling [195], Lee et al. [105], Qu et al. [168], Jiang et al. [95]	Description	Short-term	-	Closed	Nonparametric
	Conversational Health Agents [1]	Description	Short/Long-term	Empathetic	Open	Nonparametric
	Liu et al. [131]	Description	Short/Long-term	-	Open	Parametric
	CharacterChat [181]	Description	Short/Long-term	-	Open	Nonparametric
	Shea et al. [184], Li et al. [113], Be Selfish, But Wisely [25]	Dialogue	Dialogue	-	Open	Parametric
	Karra et al. [99]	Dialogue/Description	Short-term	-	Closed	Parametric
	ControlLM [227]	-	Short/Long-term	-	Open	Nonparametric

3.1 Architecture

To effectively accomplish individual simulation, it is essential to construct an agent architecture that can accurately replicate the features of the individual. Typically, this architecture is modularized into four core components: **profile**, **memory**, **planning**, and **action**.

3.1.1 Profile. Profiles define the unique characteristics of simulated individuals, including attributes, behaviors, and constraints, with variations in construction and form.

Profile Construction. Profile construction refers to the process of collecting individual-related information, categorized into *manual modification* and *LLM generation*. *Manual modification* creates high-quality profiles by integrating publicly available data under human guidance, combining coarse information from well-known characters [219, 222] and specific personalities [36, 46], with online communities contributing web-based profiles from sources like Wikipedia [183] and social media [59], while literary works add further depth by reflecting the author's thoughts [182] and character portrayals [18, 108]. *LLM generation* automatically generates persona-based profiles by prompting LLMs with individual details [176, 181, 217, 224], easily exploring diverse profiles, although quality requires careful human supervision.

Profile Form. Profile form defines the format of individual information, which can be categorized into *descriptions* and *conversations*. *Descriptions* provide basic details like name, age, and gender [89, 222], reflecting attributes without deeper context, while *conversations* reveal character profiles implicitly through dialogue, drawn from authentic sources (e.g., films, literary works, and scripts [18, 88, 112, 263]) or LLM-generated dialogues [61, 108, 224, 233, 261].

3.1.2 Memory. Memory stores perceived or generated information, enabling agents to maintain consistent behavior and overcome the limited context window of LLMs, but the complexity of memory poses challenges for researchers in designing efficient types and operations.

Memory Type. Memory can be commonly divided into two types based on its temporal nature: *short-term memory* and *long-term memory*. *Short-term memory* records the immediate perceptions of the agent, divided into the simulation contents (e.g. user instructions [46, 182], dialogue history [85, 239], and user/environment responses [242]) and the simulation supplements (e.g. scene descriptions [2, 242] and scene-related experiences [183, 247]), which navigate agents through the simulation to perform tasks appropriately. *Long-term memory* stores stable, global information to prevent deviations from goals, maintaining individual-specific data such as experiences, behaviors, knowledge, and skills [116, 247], with vector databases improving memory management, retrieval, and organization [130].

Memory Operation. Memory operations refer to the continuous updating and use of memory by the agent, including three types: *memory writing*, *memory retrieval*, and *memory reflection*. *Memory writing* involves storing relevant historical content (e.g. user-specific dialogue history [112], new skills [208], selected papers and others [218]) in memory, where useful information is retained for future use. *Memory retrieval* extracts valuable content from memory based on specific needs, with its effectiveness crucial for performance in context-sensitive simulations. Traditional retrieval methods use techniques like keyword matching [28] and embedding vectors [130], while recent works introduce the retrieval model to select the most relevant information [35, 179]. *Memory reflection*, mirroring human behavior of reconsidering past actions and thoughts, helps the agent organize [157], refine [208], and abstract memories [271]. Through memory reflection, agents can synthesize fragmented experiences into coherent knowledge, which significantly enhances performance and deepens simulation complexity, particularly in dynamic environments.

3.1.3 Planning. Planning involves deciding actions to achieve specific goals. While traditional tasks focus on solving problems like mathematical reasoning [209] or embodied tasks [188, 234], simulation agents in social

contexts must consider others' emotions and intentions while preserving internal consistency in their personality traits and beliefs. We refer to these as empathetic planning and subjective planning, respectively.

Empathetic planning. Empathetic planning involves an agent's ability to infer others' behaviors and emotions using Chain-of-Thought (CoT) reasoning, enabling adaptive decisions or judgments [107, 193, 242]. This allows the agent to tailor actions based on emotional and behavioral context, guiding personalized feedback acquisition.

Subjective planning. Subjective planning involves an agent acting based on its own thoughts and feelings, in line with its role or identity, such as fine-tuning LLMs using inner monologues [171, 183] or guiding LLMs to express beliefs through CoT [241], driven by the agent's internal state rather than external stimuli or others' needs.

3.1.4 Action. Action refers to the direct interaction between LLMs and their environment, serving as the interface for simulating human behavior and enabling tasks that replicate real-world actions and decision-making. It comprises two key aspects: the action situation and the action domain.

Action Situation. As individual simulations become more diverse and complex, various action situations emerge (e.g. dialogue [37], games [127], real world [239]), which can be divided into *simple dialogues* and *crafted situations*. *Simple dialogues* consist of few-turn conversations without restricted environments, such as interactions between characters [18], and are used to induce attributes like personality [95, 99], traits [178], and toxicity [46], as well as for persona evaluation through interviews [217] or questionnaires [153]. *Crafted situations* involve carefully designed environments with rules and contextual descriptions, often using modified games to create structured virtual spaces for users and agents, particularly in board role-playing games [102, 127]. More advanced environments, like sandboxes [28], extend beyond rules to establish objective settings, while simulations are further enhanced with elements from scripts, such as facial expressions, small movements [2, 233], and nuanced information from environmental images [44].

Action Domain. Action domain can be commonly divided into *close domain* and *open domain* based on the restriction of action space. *Closed domain* simulations involve restricted action spaces, where LLM responses are predefined by researchers for tasks like questionnaires [99], decision-making from predefined options [83], or ratings based on set standards [217], improving efficiency in practical tasks such as tool selection or recommendations [35, 264]. *Open domain* simulations allow LLMs to generate free-form responses with fewer restrictions, better mimicking real-world conditions and requiring higher standards; popular tasks include conversational actions for behavior simulation [18, 261, 289] and scenario-based interactions in environments like sandboxes [130, 208] or games [102, 127].

3.2 Construction

Construction indicates the process of integrating individual data into the established model of LLMs, which aligns the model and the individual, thus creating the simulating LLMs. Generally, construction methods are distinguished into two types, namely **nonparametric prompting** and **parametric training**.

3.2.1 Nonparametric Prompting. Nonparametric prompting, i.e., prompt engineering, is a method of interacting with LLMs by designing and optimizing input prompts. In some individual simulations, the description-based profile is implemented by a system prompt. Researchers often create system prompts that begin with "You are a..." to assign models specific demographic features and roles [46]. Besides, LLM outputs are enhanced in some works through few-shot prompting by providing examples to inject detailed information and improve response quality. Moreover, incorporating problem-specific details directly within prompt structures can significantly enhance the effectiveness of simulation.

Short-term memory is often implemented by nonparametric prompting. For situation-based individual simulations, environment descriptions and behavior rules are typically conveyed through prompt engineering [102]. Since situational information is generally objective and must be followed, emphasizing this information directly in the input is a rather effective method for constructing simulations. However, due to the context window limitations of LLMs, the quality of the profile prompt significantly restricts prompt-based individual simulations. Moreover, the preset template configurations as the “assistant” within LLMs pose a major challenge for prompt engineering in individual simulations [181].

3.2.2 Parametric Training. Parametric training modifies the model by directly updating the LLM parameters with given data. The training methods can be generally categorized into pre-training, fine-tuning, and reinforcement learning.

Pre-training. The pre-training method in individual simulation focuses on aligning the original LLMs with basic individual-related data and establishing individual knowledge for LLMs. The targets of training datasets vary in recent studies, including individual descriptions [179], literature summaries [18], and philosophical works or utterances [182].

Fine-tuning. The fine-tuning method is designed for adapting LLMs for individual simulation in specific tasks and situations. Researchers collect and modify supervised instruction datasets tailored for specific situations and fine-tune their models to equip them with the corresponding capabilities. Using persona-enhanced datasets is an effective method to regulate the models’ behavior in individual simulation, which is constructed by adding instruction tuning samples of the simulated individual’s behavior [61, 171]. LoRA fine-tuning method can integrate multiple characters into a single model [194, 261]. In multimodal finetuning scenarios, both visual and textual information are considered to significantly enhance LLMs’ simulation behavior in multimodal contexts [44, 179]. Compared to prompt engineering, fine-tuning leverages large datasets more effectively and reduces the limitations imposed by the pre-training phase of LLMs.

Reinforcement Learning. The reinforcement learning method is used to refine models in dynamic environments with the goal of maximizing cumulative rewards. In simulations involving conversations and dialogues, the quality of the LLM’s responses directly influences the rewards it receives [11, 90, 184], which encourages the model to learn the appropriate ways to respond in dialogues. By modifying the reward function, researchers can influence the model’s preference and thus manage to mimic the personas of the simulated individuals [25]. As individual simulations become more diverse and complex, reinforcement learning plays a crucial role in improving the dynamic behavior of LLMs.

3.3 Simulation Objectives

Simulation goals for individuals can be grouped into two types: (1) **Demographics**: a group of people who share the same characteristics, such as psychological traits like INTJ (Introversion, Intuition, Thinking, Judgement) or identity-related features. (2) **Characters**: a specific individual, whether real or virtual, who is widely recognized by groups of people.

3.3.1 Demographics. Demographic individuals refer to groups of people who share common attributes, such as occupation, age, or social identity. In an abstract sense, demographics can be interpreted as clustered representations in embedding space that capture shared opinions and behavioral tendencies [113]. Early studies examined how pre-trained LLMs encode demographic attributes internally [117, 131], providing a foundation for demographic-level simulations. Such simulations are often used to approximate group opinions [105], analyze collective preferences or biases [106, 168], and scale persona-conditioned interactions [38, 61, 185], thereby supporting broader societal modeling efforts [28]. In most cases, demographic simulation adopts nonparametric

prompting, where persona identities are injected into context, and carefully designed tasks or surveys are used to elicit representative behaviors [83, 214].

3.3.2 Characters. Character simulation focuses on modeling specific individuals rather than aggregate groups. These characters may include public figures, fictional roles, or platform users [108, 223, 247]. Compared to demographic simulation, character-level approaches emphasize fine-grained persona construction, narrative coherence, and behavioral consistency across interactions.

Real Characters. Real characters are typically well-documented public figures whose biographies, writings, or media presence provide structured and verifiable data sources [59, 147, 183, 203]. Their established profiles facilitate relatively objective persona construction and evaluation, as aspects of speech style, domain expertise, and historical viewpoints can be partially validated [182].

Virtual Characters. Virtual characters originate from fictional narratives in literature, film, and interactive media [26, 121, 127, 289]. Unlike real individuals, their identities are defined by narrative design rather than historical records, placing greater emphasis on internal consistency and story-aligned behavior. Virtual character simulation is frequently deployed in interactive and entertainment-oriented scenarios, where maintaining immersive dialogue and coherent personality traits is critical. However, constructing reliable persona representations for fictional roles remains challenging, particularly when datasets are sparse or inconsistently curated.

3.4 Evaluation

To measure the performance of individual simulations, provide insights into their feasibility, and guide improvements to simulation architectures, researchers have developed diverse evaluation standards and methods, ranging from simple to complex approaches. These methods can be categorized into **static evaluation** and **interactive evaluation**.

3.4.1 Static Evaluation. Static evaluation refers to the dialogue-based assessment of LLMs by directly inducing their generation and measuring their quality. It can be categorized into subjective evaluation, which involves assessments by both LLMs and human evaluators, and objective evaluation, which utilizes mathematical tools for analysis.

Subjective Evaluation. Subjective evaluation refers to assessments conducted by humans or LLMs based on subjective standards. It often involves leveraging conversations with varying forms and contexts. Interview techniques are widely adopted [217, 224] because they can effectively prompt LLMs to generate expected responses. Other approaches, such as utterance imitation [46], are also favored in some research. Once dialogues are generated, some studies utilize advanced LLMs to evaluate the output on a given scale [121, 160, 217, 261], considering performance dimensions. These dimensions range from psychology-based metrics, such as the Big Five Personality Traits (BFI) and Myers-Briggs Type Indicator (MBTI), to language-based factors like grammar and tone. Human annotators are often involved in experiments to provide human reference points [1, 9, 157].

Objective Evaluation. Objective evaluation refers to assessments based on objective indicators, utilizing mathematical and statistical tools. It takes advantage of mathematical tools to grade the generation of simulating LLMs. Examination commonly involves option choosing(or questionnaire) [99], ranking [59] and question answering [89]. Accuracy [113, 239], F1 score, recall [4] are used in option choosing and ranking. In the examination of generation(question answering), text sequence related tools such as perplexity [2, 37, 111], ROUGE-L [26, 53, 131, 239] and BLUE [59, 111, 131] are broadly used in the evaluation, especially those with a reference version [26]. Objective Examination is a more credible method of evaluating the performance of LLMs in individual simulation. However, it is highly restricted, and occasionally, specific objective tools must be developed to facilitate the evaluation of simulation in given dimensions.

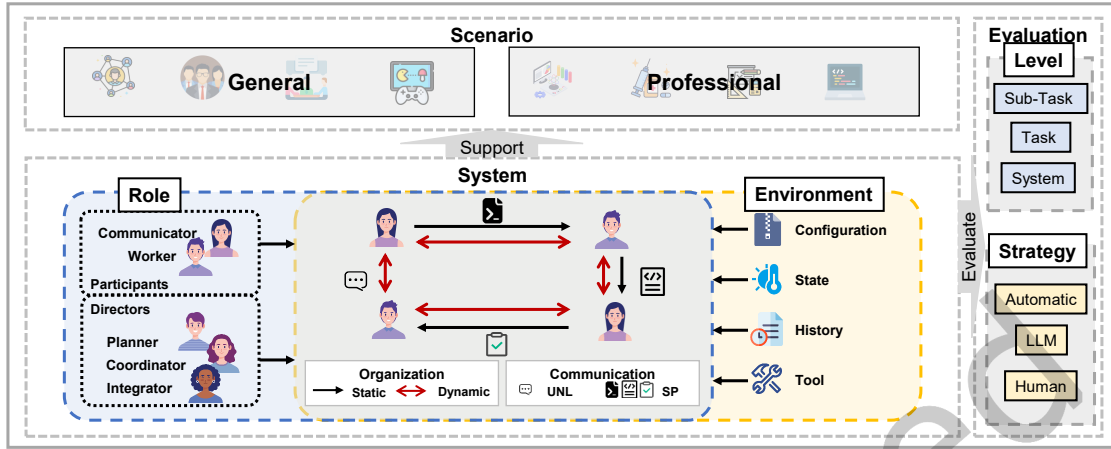


Fig. 3. Illustration of scenario simulations. Given a specific scenario, building a system involves modeling environment, roles, organization, and communication with detailed modules or mechanisms adjusted to the targeted scenario. After simulating the scenario, the desired output, typically the result of a task or problem, is obtained and evaluated using different levels and strategies.

3.4.2 Interactive Evaluation. Interactive evaluation assesses individual simulations in complex scenarios through detailed, circumstance-based interactions. It is commonly applied in game performance [102, 127], task completion [35, 52, 221], and nuanced role-playing [25, 88]. Key features include a carefully designed environment, real-time feedback, and multi-stage assessments. The crafted environment is described in §3.1.4, while real-time responses capture the environment’s reaction to the agent’s outputs. Interactions between agents and the environment generate multi-turn dialogues, revealing LLMs’ capabilities in dynamic contexts. Because single-aspect metrics are insufficient, studies often evaluate a range of objectives—from specific actions to hybrid behaviors [218], or from single-turn to multi-turn interactions [183]. Other work assesses generation quality, including accuracy, tone imitation [85, 224], and self-consistency [135]. Overall, interactive evaluation emphasizes not only correctness but also how closely simulations resemble real-world scenarios.

3.5 Discussion

Comparison of Simulation Schemas. Table 2 systematically compares different individual simulation paradigms. Parameter-embedded methods fine-tune model parameters to encode persona, achieving high behavioral consistency but requiring significant data and retraining, which limits scalability. Context-compressed methods inject structured persona summaries during inference, avoiding retraining but risking information loss or drift over long interactions. Conditional-control methods use prompts or labels to guide generation without updating model weights, enabling scalable and low-cost deployment, though often with shallower persona modeling and potential stereotyping. In practice, the choice depends on whether the goal prioritizes depth of behavior, adaptability in context, or breadth of coverage, while being mindful of each paradigm’s characteristic limitations.

Reflection on Evaluation. Evaluation remains one of the central bottlenecks in individual-level simulation, primarily due to challenges of reliability and replicability. Current objective evaluations are largely confined to coarse-grained metrics or static multiple-choice style assessments, which are particularly ineffective in open-domain dialogue settings. As a result, subjective evaluation by humans or LLMs is still widely used. However, such approaches are costly, lack replicability, and make it difficult to ascertain whether different methods truly yield improvements. Moreover, existing efforts predominantly focus on isolated, short-term scenarios, while lifelong

Table 2. Comparative analysis of individual simulation paradigms: fidelity, scalability, cost, transferability and typical failure modes.

Paradigm	Representative Papers	Core Mechanism	Fidelity	Scalability	Cost	Transferability	Failure
Parameter-Embedded	CharacterLLM [183] Capturing Minds [171]	Role-specific fine-tuning	High	Low	High	Limited	Overfitting, catastrophic forgetting, hallucinated persona
	Conversational Health Agents [1] Shea et al. [184]	Assistant-oriented fine-tuning	Moderate-High	Low	High	Moderate	Task over-optimization, persona dilution
Context-Compressed	InCharacter [217] LifeChoice [247]	Persona summarization and injection	Moderate	Moderate	Moderate	Moderate	Information loss, drift, noisy summaries
	ControlLM [227] Marked Personas [36]	Mechanism-based control	Weak-Moderate	High	Moderate	Moderate	Control instability, persona entanglement
Conditional-Control	PersonaHub [61] CharacterChat [181]	Diverse persona prompting	Moderate	High	Low	Broad but shallow	Stereotype amplification, out-of-character behavior
	Out-of-One-Many [6] Lee et al. [105]	Label combination	Weak	Moderate-High	Low	Moderate	Label conflict, inconsistent persona composition

evaluation, such as assessing the long-term consistency of behavioral roles, remains largely absent. Moving forward, there is a pressing need to establish standardized annotation protocols, to incorporate the performance of simulated individuals in downstream tasks such as recommendation or interviews as an indirect measure of simulation effectiveness, and to develop lifelong evaluation paradigms that can capture long-term consistency and coherence of individual behaviors.

4 Scenario Simulation

Table 3. A list of representative works of scenario simulation. UNL: unstructured natural language; SL: structured language.

Scenario	Task	Paper	Environment			Director	Organization	Communication
			Configuration	State	History			
Dialog-Driven	Social Interaction	Sotopia [291]	✓	✓	✓		static	UNL
		APAM [254], SimuLife++ [253], Elicitron [7]	✓		✓		static	UNL
		Self-Emotion [277]	✓		✓	✓	dynamic	UNL
	Question Answering	ICL-AIF [55], MAD [125], ChatEval [23], FORD [245]	✓		✓	✓	static	UNL
		Du et al. [49], AmazonHistoryPrice [238]	✓		✓		static	UNL
		AutoGen [231], DoG [139]	✓		✓		dynamic	UNL
	Game	ChatLLM [75]	✓				static	UNL
		Xu et al. [249], ReCon [216], Lan et al. [103], Xu et al. [251]	✓		✓		static	UNL
		MachineSoM [275], ThinkThrive [230], PLAYER [294]	✓		✓		dynamic	UNL
		WWQA [48]	✓		✓	✓	static	UNL
		GITM [295]	✓	✓	✓	✓	static	UNL
		Sreedhar et al. [190]	✓	✓	✓		static	UNL
Task-Driven	Foundational and Applied Science	S-Agents [30]	✓	✓	✓	✓	dynamic	UNL
		VIDS [77]	✓		✓		dynamic	UNL
		ChatGPT Research Group [288]	✓	✓	✓	✓	dynamic	UNL
		MARG [45], DERA [151]	✓		✓	✓	static	UNL
		AI Hospital [51], DreamFactory [244]	✓		✓	✓	static	SL
		REVIEWER2 [60], ResearchAgent [10]	✓				static	UNL
		CosmoAgent [96], CulturePark [109], SynthPAI [267]	✓		✓	✓	dynamic	UNL
		Self-collaboration [47]	✓			✓	dynamic	UNL
	Software Development	ChatDev [162]	✓	✓	✓		static	SL
		MetaGPT [82]	✓	✓	✓	✓	static	SL
		Experiential Co-Learning [163]	✓	✓	✓		dynamic	SL
		AutoCodeRover [282]	✓		✓		static	SL
		IER [164]	✓		✓		dynamic	SL
		TradingGPT [120], Multi-Agent Collaboration [197]	✓	✓	✓		dynamic	UNL
	Other Industries	Information Bazaar [226], Baker et al. [12]	✓		✓		static	UNL
		MATHVC [266], CAMEL [110]	✓		✓	✓	static	UNL
		LawLuo [192], AutoAgents [27]	✓		✓	✓	dynamic	UNL
		MAIC [259], AgentVerse [31], Scalable [34], ReAd [283]	✓	✓	✓	✓	dynamic	UNL
		SwiftSage [129], RoCo [140], CoELA [272]	✓	✓	✓		static	UNL

In real world, individuals do not function separately. Instead, they frequently engage in collaboration to complete tasks within specific scenarios. This raises a key question: can agents cooperate like humans, or even surpass human performance in achieving collective intelligence? To explore this, researchers simulate multi-agent collaborations across diverse scenarios. A scenario simulation typically starts with designing a multi-agent system that includes constructing the scenario environment, modeling agent roles, and establishing organizational structures and communication protocols to manage agent interaction. In this section, we begin discussing the system composition of a scenario simulation in §4.1. Following this, we summarize several scenarios that have recently attracted the attention of researchers in §4.2. Finally, we review the methods and metrics commonly used for evaluation in §4.3 and present critical discussions in §4.4. The overall framework is presented in Figure 3 and representative works are summarized in Table 3.

4.1 System

The diversity of scenarios presents challenges in proposing a unified system applicable to scenarios. Most current systems can be summarized as “*agents organized to play different roles in dedicated environments through restricted communications*”. Based on this general description, we identify four key concepts in scenario simulations: **environment**, **role**, **organization**, and **communication**.

4.1.1 Environment. The environment in scenario simulation defines the specific contexts in which agents operate and interact with each other. Like we humans, agents depend on the environment to receive input from various sources. These signals guide the behaviors and strategies of agents within the system. We analyze the environment of existing work by focusing on three key aspects: *configuration*, *state*, and *history*.

Configuration. The environment configuration provides basic information, especially essential elements necessary for the tasks and goals in the scenario. The system will initialize agents accordingly so that they interact with clear objectives. A typical environment configuration may include *events* in the environment, such as specific court cases [12, 72, 78, 192], and topics for multi-agent debates [23, 49, 125, 139, 231, 238, 245]. The *profile* of agents is also a key component of the configuration, which encompasses agents’ interests, goals, roles, etc. [82, 267, 277]. Agents may also have access to *external resources*, such as related papers [10], predefined strategies [277] or disease information [114].

State. While configurations provide information *before* the scenario, environment states record instant information from the environment *during* the scenario. They directly influence the agents’ decision-making and behavior. For example, *observation* involves changes in the environment and the current state of surrounding entities, such as the properties of other agents [30, 34, 129, 198] and agents’ physical states [34, 198, 283, 295]. On the other hand, *feedback* consists of responses received by agents after their actions. Feedback can manipulate agents’ strategies and even cognitive states to simulate human-like adaptability [30, 190, 197]. Feedback may also come from events [120], others’ decisions [190] or external tools [82, 162, 231].

History. As the scenario runs, past states and interactions accumulate into historical records. Agents can leverage them to adapt to new situations and refine strategies, ensuring more coherent and effective task performance in dynamic environments. The most direct way is to append the history (e.g. past dialogue) to the current input without modification [49, 125, 229, 231]. Excess content may be truncated to limit the token while preserving key historical information [34, 243]. Further exploitation is also common, for example, history-based iterative refinement [45, 139, 226] and dedicated summarization [162, 200, 295]. A more complex approach is to use agents’ memories to process the history, enabling agents to preserve relevant information both within and across sessions [27, 82, 120, 130, 164, 165, 244].

4.1.2 Role. In scenario simulations, we assign agents distinct roles based on their tasks and functionalities. As demonstrated in Figure 3, there are three groups of roles in a typical setting: *communicators* exchange information

between agents and the environment (e.g. human users), *workers* carry out the tasks within the scenario, and (optional) *directors* manage the task execution processes while providing necessary assistance. Each role has its own responsibility that emphasizes different aspects of the system's operations. They collaborate to achieve the system's overall goals.

Communicators. Communicators primarily focus on communication, such as information exchange, feedback, and task guidance. Specifically, this kind of agent can process information for certain disciplines and research applications [72, 151] and advocate diverse viewpoints [75, 245], claims [125] and underlying needs [7, 109].

Workers. Workers are directly involved in task execution and operations, demonstrating specialized skills and efficiency. This typically includes the common professional roles present in each scenario, such as coder and tester in software development [47], buyer and seller in negotiations [55], doctors and medical professional agents in the healthcare domain [114, 229], and receptionist, lawyer, and secretary in legal contexts [192]. Workers can also be equipped with external tools with specialized functionalities to enable more accurate and precise outcomes, ranging from programming languages such as Python [82, 162, 231] and SQL [10, 139, 293] to external APIs [27, 243].

Directors. Directors provide essential support indirectly through various crucial aspects. For example, planners define tasks and formulate strategies for effective inter-agent collaboration; they may define objectives [31, 47, 110, 295], analyze user requirements [82, 293] and optimize execution plans [30, 34]. Coordinators manage the collaboration between agents to ensure effective task execution [78, 82, 96, 109, 288], monitor progress [125, 245] and facilitate cooperation [55, 123]. Finally, integrators encompass various decision-making and summarization functions critical to guide the system's trajectory [23, 51, 78, 151, 200].

4.1.3 Organization. Effective task execution necessitates careful coordination and scheduling of the interactions between individual agents. The organizational structures establish how each agent collaborates with others to achieve a goal. Typically, we can depict an organization schema by its *mode* and *structure*.

Mode. The organizational structure determines whether the agent relationship remains stable or evolves dynamically throughout the simulation. In terms of how to organize agents, there are mainly two modes in existing research: *static* and *dynamic*. *Static mode* refers to the organizational structure predefined based on the nature of the tasks. Agents communicate and work in an orderly manner according to these static structures, such as structured debates [23, 55, 110, 151], skill training [253, 254] and integrating ideas [72, 75]. The multi-stage version divides tasks into distinct stages and changes organization between them, which can be found in the design, coding, and testing stages in software development scenarios [82, 162], as well as multi-stage process in judicial scenarios [12, 78] and problem-solving processes [139, 272, 295]. *Dynamic mode* explores more open and adaptive organizational structures, often relying on dynamic and heuristic communication. Agents can be flexibly created and recruited [27, 31, 134, 139, 243], coordinated through liaison agents [96, 109], and self-organized [30]. This mode can also be multi-staged, where agents across stages communicate autonomously based on the current state [47, 192, 200, 259, 288].

Structure. The organization structure, meanwhile, reflects how agents are connected. Typically, an organization can be *layered*, *centralized*, or *decentralized*. *Layered* structures adopt a hierarchical framework that assigns agents to distinct levels. Interactions are predominantly confined to agents within the same level or between adjacent layers, facilitating a controlled and organized flow of information [72, 75, 162]. *Centralized* structures often involve a high-level role (e.g., coordinator) that serves as the core of the organization, overseeing communication and functioning as the central hub for interactions among other agents [51, 96, 109]. *Decentralized* structures, in contrast, are more flattened, where agents can engage in peer-to-peer interactions as needed [23, 125, 139].

4.1.4 Communication. The communication between agents controls the transmission of information. To better understand the internal communication mechanism, we dissect communication from its *format* and *style*.

Format. From the perspective of information format, there are two common communication protocols: *unstructured natural language* and *structured language*. *Unstructured natural language* is most commonly used in multi-agent communication, enabling flexible and immediate exchanges through free-form conversational language that mirrors human dialogue [49, 55, 110, 151, 245, 253, 254, 288]. Yet, this format suffers from its ambiguity and redundancy, where *structured language* becomes more helpful, such as code and JSON documents. For example, code is used to pass information between phases in software development [82, 162], while structured summaries are utilized to gain key insights in medical reports [200]. In addition, agents can also autonomously choose the appropriate format during interactions to improve efficiency [32, 33]. Recent studies have also attempted to involve multiple languages [141].

Style. Communication, by nature, can be *cooperative* or *competitive* regarding its style. In *cooperative* communication, agents share a common objective, aiming to optimize collective outcomes, like software development [47, 82, 162], medical diagnosis [51, 200], and case handling [72, 192]. In contrast, agents in *competitive* communication typically hold differing viewpoints and positions, each striving to achieve individual objectives. Such scenarios are commonly found in settings like games [48, 216, 249] and debates [55, 125, 245], where agents maintain opposing stances and seek to outmaneuver each other.

4.2 Scenario

With the collective capabilities of agents with expertise, scenario simulations have been applied to various domains. Here we divide different scenarios into two groups: **dialog-driven** ones covering social interaction and question-answering, and **task-driven** ones focusing on specialized tasks.

4.2.1 Dialog-Driven Scenario. Dialog-driven scenarios are the ones in people's daily lives where the dialog is centered, such as those for social or entertainment purposes. They share a common emphasis on tackling general goals unrelated to any specific task or domain. Examples include *social interaction*, *question-answering*, and *games*.

Social Interaction. To model realistic interpersonal dynamics, frameworks like Sotopia [291] and SimuLife++ [253] provide environments for evaluating social intelligence. Recent advancements enhance fidelity through objective assessments [207] and interactive learning [215, 254], enabling agents to dynamically acquire social norms [278, 290] and reason about others' mental states [146].

Question Answering. In multi-agent settings, question answering evolves into a collaborative reasoning process. To mitigate hallucinations, researchers employ debate mechanisms where agents critique each other's outputs [245] or operate under judicial supervision [49]. Beyond factual accuracy, frameworks simulate negotiation [238] and utilize "society of minds" architectures [49] to optimize strategic decision-making and consensus.

Game. Games serve as high-fidelity testbeds for studying strategic behavior under uncertainty. Classic environments like Werewolf and Avalon are widely employed to quantify abstract concepts such as trust and deception [216, 249, 258]. To master these complex dynamics, agents are augmented with reinforcement learning for strategy adaptation [191, 232, 251] or integrated with external tools for logical deduction [294]. Furthermore, these settings are pivotal for investigating ad hoc teamwork, where agents coordinate without prior protocols [63, 187, 230].

4.2.2 Task-Driven Scenario. In task-driven scenarios, agents role-play personas with specific functions for certain tasks. Most fall into one or more specific domains related to the objective tasks. Agents are increasingly leveraged to solve complex, domain-specific problems by automating tasks and improving decision-making processes.

Foundational and Applied Science. Multi-agent systems act as force multipliers in scientific discovery, streamlining both experimental and academic workflows. In experimental fields, agents collaborate to plan syntheses and refine diagnoses, frequently integrating with Bayesian optimization or RAG tools to ensure rigor [114, 200, 201, 229, 288]. Complementing this experimental work, agents also automate scholarly tasks, including literature reviews [175], presentation generation [286], and data visualization [77].

Software Development. Representing a mature application of Structured Language, frameworks such as ChatDev [162] and MetaGPT [82] simulate virtual software companies by enforcing SOPs across engineering roles. This "assembly line" approach ensures code quality through iterative verification. To enhance adaptability, recent research has shifted towards self-evolutionary systems, where agents learn from debugging history [163, 164, 250] or autonomously resolve repository-level issues [282].

Other Industries. Simulations are also reshaping decision-making protocols in broad professional sectors. Systems model legal trials to predict outcomes [78, 192], employ layered memory for financial trading and market analysis [120, 226, 246], or emulate classroom dynamics to support pedagogical research [259, 266].

4.3 Evaluation

For scenario simulations, the evaluation focuses on how well the tasks of the scenarios are solved. Based on the scope of the evaluation, it can be categorized into **task evaluation**, **sub-task evaluation** and **system evaluation**, each employing various *automatic*, *LLM-based*, and *human* evaluation methods to assess performance.

Task Evaluation. Task Evaluation measures the overall performance of tasks assigned to the scenario. The evaluation can be carried out in automatic ways or by LLMs or humans. In terms of *automatic* evaluation, predefined metrics and mathematical tools are used to objectively assess the task outcomes, such as accuracy [72, 245], pass@k [110] for coding tasks, success rate, and coverage for exploration [295], and deal price for negotiation [55]. These methods are efficient and scalable, but may overlook complex behaviors. Thus, *LLMs* [75] and *human* experts [110, 125] have been applied to provide more nuanced evaluation for qualitative tasks and compare solutions based on specific criteria.

Sub-Task Evaluation. Sub-task Evaluation assesses the completion of sub-tasks within a scenario simulation and their impact on overall task performance. It serves as a process evaluation for the execution of complex tasks. The *automatic* evaluation uses metrics like transport rate, average steps, task success rate, re-plan attempts, and efficiency improvement to assess sub-task performance and strategy efficiency [140, 272]. Completeness, executability, and consistency metrics are often applied in software generation tasks [162, 163]. *LLM-based* evaluation focuses on pairwise comparisons or win rate judgments, capturing qualitative aspects of sub-task performance [162]. Meanwhile, *human* evaluation relies on participants to provide subjective assessments on metrics such as executability, revision costs, or comment quality, offering practical insight into sub-task performance [45, 82].

System Evaluation. System Evaluation aims to capture the effectiveness and efficiency of the system as a whole. *Automatic* evaluation relies on metrics such as token consumption, task win rate, and human-likeness scores to measure the efficiency and realism of agents [198]. Additional metrics like accuracy, precision, recall, and F1 scores are used to assess system accuracy and consistency in diagnostic or predictive tasks [51]. *LLM-based* evaluation often involves GPT-4 to assess qualitative aspects, such as human-likeness or diagnostic report quality [114, 198]. *Human* evaluation typically involves subjective assessments, such as rating instructional content for tone, clarity, and supportiveness on a Likert scale [259], often used to complement automatic methods and capture human perspectives on system outputs.

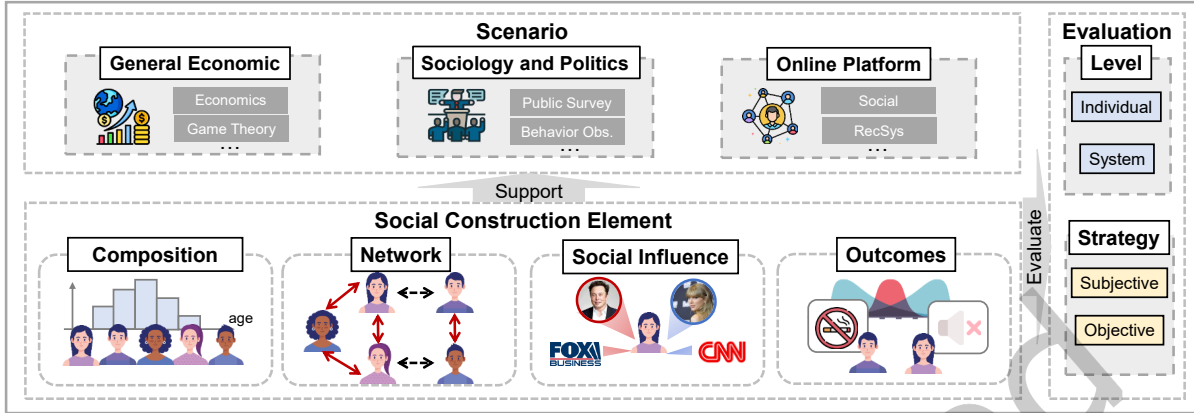


Fig. 4. Illustration of society simulations. To construct society simulations, the corresponding society’s construction elements, i.e., composition, network, social influence and outcomes need to be carefully designed. Building on this, various scenarios can be simulated. The performance of individuals and the overall performance of the system are evaluated.

Table 4. Comparative analysis of scenario simulation architectures: determinism, stability, latency, cost and typical failure situations.

Architecture	Representative Papers	Core Mechanism	Determinism	Stability	Latency	Cost	Failure
Static Pipeline	ChatDev [162], MetaGPT [82]	Predefined sequential flows or DAGs	High	Low	Low	Low	Error Propagation, Schema Rigidity
Unstructured Debate	MAD [125], ChatEval [23]	Free-form consensus discussion	Low	Med	High	High	Social Sycophancy, Context Dilution
Dynamic Organization	AgentVerse [31], AutoGen [231]	Runtime topology evolution	Med	Low	High	Med	Planning Bottleneck, Complexity Overhead
Environment-Centric	GITM [295], Agent Hospital [51]	Environment-driven feedback	High	High	Med	High	State Desync, Infinite Loops

4.4 Discussion

Comparison of Simulation Schemas. As summarized in Table 4, scenario simulation architectures can be categorized into four paradigms based on their mechanisms and performance trade-offs. Static Pipeline (SOP) approaches [82, 162] enforce sequential or DAG-based execution to ensure high determinism and low latency, though they are structurally brittle, prone to error propagation and schema rigidity when handling edge cases. In contrast, Unstructured Debate frameworks [23, 125] prioritize dialectical reasoning to foster consensus, yet this comes at the expense of high latency and token costs, with frequent risks of social sycophancy and context dilution. Bridging these extremes, Dynamic Organization architectures [31, 231] adaptively evolve agent topology at runtime, offering a balance in costs but introducing planning bottlenecks and complexity overhead. Finally, Environment-Centric systems [51, 295] drive interaction via external feedback to achieve high stability and fidelity; however, this integration incurs high costs and susceptibility to state desynchronization or infinite loops. Ultimately, the choice of architecture requires balancing deterministic efficiency against the robustness of emergent behaviors.

Reflection on Evaluation. In task-driven scenario, evaluation can rely on objective metrics, including correctness, efficiency, or token consumption. By contrast, dialogue-based scenarios remain challenging, as assessment still depends heavily on human and LLM-based judgments. Some studies [291] suggest that LLM-based evaluations can align reasonably well with human assessments on goal completion, but substantial gaps remain in other dimensions. These limitations highlight the trade-offs of current approaches: human evaluation is costly and subjective, while LLM-based evaluation may be biased or unreliable. Future work should aim to narrow this reliability gap and develop fine-grained metrics that capture both efficiency and qualitative aspects. In domain-specific contexts, expert input is often needed to identify the most relevant evaluation dimensions, encompassing

Table 5. A list of representative works of society simulation.

Scenario	Field	Paper	# Agents	Construction Element			
				Composition	Network	Social Influence	Outcome
General Economic	Game Theory and Strategic Interactions	Agent-trust [241]	(0, 10]	✓		✓	✓
		Economics Arena [69], Fontana et al. [54], LELMA [142], Mozikov et al. [150]	(0, 10]			✓	✓
		SABM [74], CompeteAI [285], WarAgent [84]	(0, 10]	✓	✓	✓	✓
	Economic Contexts	Horton [83], EconAgent [116]	(10, 100]	✓			✓
		SRAP-Agent [92], Williams et al. [228]	(10, 100]	✓	✓	✓	✓
		AgentTorch [39]	(100, ∞)	✓	✓		✓
Sociology and Political Science	Public Opinion Survey	Argyle et al. [6], ElectionSim [279]	(100, ∞)	✓			✓
		Park et al. [158], Sun et al. [195]					
		GABSS [240]	(100, ∞)	✓	✓	✓	✓
	Individual and Organizational Behavior Observation	Aher et al. [3]	(0, 10]			✓	✓
		Zhang et al. [275], Jarrett et al. [91]	(0, 10]	✓			✓
		Lyfe Agents [98], CRSEC [173]					
		Chuang et al. [42], ChoiceMates [155]	(0, 10]	✓	✓	✓	✓
		AgentReview [97]	(0, 10]	✓	✓		✓
		Generative Agents [157], AGA [262]					
		MineLand [260], CareerAgent [292]	(10, 100]	✓	✓	✓	✓
		Suzuki and Arita [196]	(10, 100]	✓		✓	✓
Online Platform	Social Platforms	Chuang et al. [41]	(100, ∞)	✓			✓
		MATRIX [199]	(100, ∞)	✓	✓		
		Cai et al. [19]	(0, 10]			✓	✓
		FPS [137], FUSE [136], Wang et al. [206]	(10, 100]	✓	✓	✓	✓
	Recommendation Environments	Social Simulacra [156], S ³ [56]	(100, ∞)	✓	✓	✓	✓
		HiSim [148], OASIS [256], BASES [172]					
		InteRecAgent [87]	(0, 10]	✓			
		Rec4Agentverse [276]	(0, 10]	✓	✓		
		RecAgent [210]	(10, 100]	✓	✓	✓	✓
		Agent4Rec [270], AgentCF [274]	(100, ∞)	✓	✓	✓	✓

process- and outcome-oriented criteria. LLMs may assist in this process, helping to define metrics and guide more systematic assessments.

5 Society Simulation

While scenario simulations focus on multi-agent interactions within small-scale, domain-specific contexts, society is far more complex, involving diverse components, varied structures, and nonlinear effects [189]. Society simulation, in contrast, aims to investigate societal dynamics and emergent behaviors rather than solving specific tasks. It serves as a vital tool for theoretical validation and predicting social dynamics. This section first summarizes the components of social construction relevant to society simulations in §5.1, followed by an overview of different scenario categories in §5.2, and concludes with a discussion of evaluation methods in §5.3 and critical discussions in §5.4. The overall framework is shown in Figure 4, and key works are summarized in Table 5.

5.1 Social Construction Elements

Considering the complexity of society, a major challenge in society simulation is bridging the gap between individual and societal scales. To address this, we identify four interrelated elements that underpin societal structures and dynamics. First, **composition** characterizes the attributes and diversity of individuals, and **network** further describes the connections and relations among them, which shape the patterns of interactions. The interactions give rise to **social influence** on individuals, leading to **outcomes**, i.e., the collective results and emergent phenomena.

5.1.1 Composition. Society is composed of massive and diverse individuals. This diversity, also referred to as heterogeneity [189] in social science, encompasses a wide range of beliefs, preferences, behaviors, normative

values, and positions within social structures. Accurately simulating societal dynamics requires carefully designing individual composition, which defines the population's diversity and realism. When modeling individual composition, we may face challenges in balancing fidelity and scalability and addressing the role of outliers.

Individual Composition. To model a diverse society, the composition of individuals in society needs to be determined. There are three main approaches to determine the composition of individuals in a system simulating a microcosm of society. Some works rely on *virtual individual synthesis*, often not focused on alignment with the real world, aiming to ensure that the system includes users with various attributes, typically by generating virtual individuals with the help of LLMs or humans [14, 43]. Other works utilize *existing datasets*, such as MovieLens-1M [210, 270], to define user composition within a simulated recommendation platform. Agents are initialized based on the user information within these datasets, reflecting the distribution of users in that context. Recently, an increasing number of studies have focused on *real-world distribution replication*, such as the composition of users on social platforms [256] or the distribution of voters in surveys [279]. For small-scale individual sets, individual data are typically collected manually [155, 158]. In cases where large-scale populations are required or obtaining real data is difficult, individuals may be sampled based on real-world macro distributions or generated by LLMs to match desired attribute distribution [6, 105, 279].

Trade-off between Simulation Precision and Scale. A key challenge in modeling composition lies in balancing the simulation precision and scale. When simulating individuals, many studies adopt detailed role modeling to enhance the authenticity of agent behavior. Beyond common demographic attributes, this may include factors such as an individual's past statements and interaction history [54, 157, 210, 270, 285]. However, as the number of individuals increases, such fine-grained modeling becomes expensive. Consequently, a trade-off often arises between the precision of individual modeling and the scale of the simulation. In large-scale simulations, to reduce computational costs, the details of each agent are typically simplified, by retaining only the most essential and common characteristics [39, 228], compressing auxiliary dialogue information into shared memory [262] or simplifying the communication [149].

Special Modeling on Outliers. While most simulations aim to capture the average population, outliers, i.e., individuals with highly distinct attributes or behaviors, often exert a significant influence in shaping social dynamics. For instance, celebrities and opinion leaders frequently occupy central positions in social networks and exert amplified influence on others [248, 280], who frequently hold prominent positions within social structures and amplify influence. In situations with limited resources, some studies [148] prioritize detailed modeling of these core content producers, while simplifying the modeling for the majority. Meanwhile, intervention policies based on simulation results often focus on these key nodes in networks [118], aiming to influence the overall system's behavior by blocking or interfering with them.

5.1.2 Network. Social interactions often occur through networks, which can be represented as graphs, with nodes symbolizing individuals and edges denoting their relationships. These networks shape the pathways of information and influence dissemination. Broadly, we categorize these networks into offline networks and online networks based on their contexts and interaction modalities.

Offline Network. An offline network represents connections formed through in-person interactions, such as face-to-face communication or the spread of diseases. On one hand, some studies aim to simulate interactions in virtual worlds, thus determining the connections between agents in a *random or predefined* manner [157, 173, 262]. On the other hand, while some studies aim to simulate the spread of a disease or event information in the real world, considering the difficulty of obtaining real data, they often estimate the social relations using *external algorithms* or agents themselves [16, 152, 228, 240]. However, in studies with massive agents, the network relationships between individuals are sometimes ignored, and individuals are treated as independent [279]. Alternatively, some

studies provide rough information, such as community statistics, in place of specific details about the agents' neighbors [39].

Online Network. An online network is a digital structure where individuals or entities interact through platforms, such as online social platforms and recommendation platforms, forming connections based on activities, relationships, or shared interests. At the beginning, some studies *randomly* initialize the social relations for users in existing datasets [210] or synthesized users [137], while other efforts have focused on crawling *authentic* social relationships from social media platforms like Weibo [56] and Twitter [148]. However, as the scale of individuals increases, it may be challenging to obtain all of their authentic relationships. Therefore, some studies construct networks using a small portion of real relationship data combined with a large amount of *synthetic* relationship data [256], or connect similar users based on the assumption of homophily [199].

5.1.3 Social Influence. Social influence refers to the process through which individuals' thoughts, decisions and behaviors are shaped by others within a social environment. It operates through various mechanisms, such as direct communication and implicit group pressure. In simulations, social influence can be discussed from two perspectives: influence received and influence exerted.

Influence Received by the Influencee. The same information may produce different effects when received by individuals with different traits. Currently, most studies have modeled how the influence received by the recipient varies based on their profile [56, 137, 256]. This can be easily achieved by integrating the individual's profile, memory and information received from others into the same context. Building on this, some works further induce additional mechanisms like cognitive bias [42] and reflection on norms [173] to enhance agents' understanding and perception of the messages.

Influence Exerted by the Influencer. The same message can have different social impacts depending on who delivers it. Studies show that influence and attention often concentrate on a small group of dominant individuals, consistent with the Pareto distribution and Matthew Effect [148, 210]. Accordingly, simulations must account for the sender's identity, status, and reputation. Some approaches model opinion leaders directly using real-world data [248, 280], while others capture influence indirectly through relational factors such as social memory [262] or party affiliation [43]. Beyond individual effects, research shows that as group size grows, the impact of any single influencer may decline, while collective influence often drives individuals toward conformity, producing herd effects [256].

5.1.4 Outcomes. Social emergence suggests that the collective behaviors or phenomena arising from individual interactions are not a linear sum of individual actions but rather complex patterns emerge from the interactions [180, 189]. The emergent outcomes can either be measurable macro results, such as voting results and public opinion levels, or qualitative social phenomena and norms.

Macro Statistical Results. Macro statistical results are typically the focus of existing studies, as they are closely related to predefined research objectives such as market research, election predictions, and public opinion forecasting. These studies often aim to calculate the sum or average of the choices or opinions of all agents in the system. To get a static opinion distribution, some studies overlook the social interactions and instead directly sum up individual choices to obtain macro outcomes [195, 279], simplifying the complexity of social dynamics. Another line of research focuses on the change of indicators by modeling multiple rounds of interactions among the agents over a period of time and then statistically analyzing the results [56, 74, 116, 202, 235].

Formation of Social Phenomena and Social Norms. In addition to the quantifiable macro results, some social phenomena and social norms are also important outcomes of social interactions. On the one hand, some studies have identified the bubble effect in recommendation systems [270], echo chambers in social media [148, 206, 256], Matthew effect in competitive agent interactions [285], and spontaneous cooperation of competing agents [235]

by calculating additional metrics or observing the trends of primary indicators. On the other hand, some studies examine social norms as an important byproduct of social interactions. This includes simulating and testing whether community rules can shape desired social norms [156], constructing normative architecture to observe the emergence of social norms [173], studying how social media language evolves in the presence of regulatory constraints [19], and observing changes in social norms in real-world scenarios such as autonomous driving [205].

5.2 Scenario

Society simulation has been widely applied to various scenarios related to human society. These scenarios cover different aspects of daily human life, and existing studies can be categorized into three primary areas: **general economics, sociology and political science**, as well as **online platforms**.

5.2.1 General Economics. Simulations in general economics analyze decision-making and behaviors related to resource allocation and competition. These studies primarily investigate how agents behave under the influence of economic incentives, market rules and resource constraints, while examining how interactions among groups shape broader economic trends.

Game Theory and Strategic Interactions. Some research focuses on game theory and strategic interaction, often involving small groups of agents and complex interactions. Classic games like the Prisoner's Dilemma are used to explore agent behaviors such as trust, decision-making, rationality, cooperation, and the impact of emotions on rational choices [21, 54, 69, 94, 142, 150, 161, 212, 241]. Other studies examine real-world scenarios like spontaneous cooperation [235], market behaviors in firm competition [74], and competition between restaurants and customers [285]. While game-theoretic scenarios offer controlled settings for studying agent behavior, real-world scenarios provide deeper insights into practical applications.

Economic Contexts. In addition to close studies on game theory and strategic interactions, some studies focus on the use of agents and their interactions within economic environments. Horton [83] examines economic agents driven by LLMs in various experiments to replicate human behavior in economic scenarios. EconAgent [116] introduces agents for macroeconomic simulation, emphasizing the influence of macroeconomic trends. SRAP-Agent [92] proposes a framework for simulating and optimizing scarce resource allocation in economics, specifically in public housing allocation scenarios. Besides, some studies involve broader macroeconomic domains, using agents to simulate and predict the spread of diseases and the change in unemployment rates [39, 228].

5.2.2 Sociology and Political Science. Society simulation has been widely used in sociological and political science research. These studies range from small-scale laboratory experiments that validate theories and hypotheses to large-scale social surveys aimed at understanding public choices. The goal is to leverage agents as substitutes for humans in studying human behavior within sociological and political contexts.

Public Opinion Survey. A key application of society simulation is public opinion surveys, which predict group perspectives on a given topic to support needs like election forecasting and public administration. Argyle et al.[6] first proposed using LLMs as silicon samples of humans in large-scale surveys. Upon this, studies have explored opinion survey scenarios such as election polls[279] and responses to public crises [240], addressing issues like population complexity and algorithmic bias. Recently, agents have demonstrated the potential to replicate participants' responses in interviews [158]. These studies lay the foundation for new tools to investigate individual and collective behavior.

Individual and Organizational Behavior Observation. Other studies focus on observing individual or organizational behavior in common or specific settings. Some works do not specify a particular scenario but instead observe agents' social interactions and potential phenomena in daily life within a sandbox environment [98, 157, 159, 173, 260]. Other studies aim to validate theories or hypotheses in specific scenarios, such as

the wisdom of partisan crowds [43], information management [155], organizational behavior management [292], and the evolution of personality traits [196].

5.2.3 Online Platform. Online Platforms are a vital component of society simulation, offering a practical means to study complex social phenomena in digital environments. These platforms, ranging from social media to online communities, allow agents to simulate real-world interactions and study dynamics such as opinion formation, information spread, and collective behaviors.

Social Platforms. Online social platforms like X (former Twitter) and Reddit have long served as an important testing ground for studying the propagation of information and the evolution of opinions. Researchers simulate behaviors such as sharing, commenting, and liking to model the dissemination of information and shifts in user attitudes on topics like news, rumors, and the influence of opinion leaders [19, 56, 66, 136, 137, 167, 174, 280, 287]. Early studies relied on real-world data, but as simulations grew in scale, synthetic data became more common [256]. To simulate larger agent populations, some studies have developed platforms that use parallel processing to improve efficiency [58, 154].

Recommendation Environments. Another widely studied scenario is the recommendation environment, where agents simulate user responses to validate and enhance recommendation algorithms [87, 276]. These studies often model personalized behaviors like item selection, preferences, and emotional responses, incorporating factors such as user memory and context [133, 210, 270, 274]. Some approaches also integrate external knowledge or self-reflection mechanisms, enabling agents to learn from interactions [220]. Together, these studies demonstrate how LLMs bridge the gap between traditional recommender systems and more human-like behavior simulations, improving accuracy and understanding user dynamics.

5.3 Evaluation

For society simulations, the evaluation primarily focuses on the comparison between the simulation results and real-world data, with assessments centered on micro level, macro level and system level.

Micro-level Evaluation. Individual simulation accuracy is key to society simulation. Therefore, micro-level evaluation of society simulation has received widespread attention. Initially, evaluations in non-real-world simulations draw on the Turing test, assessing agent behavior's resemblance to human behavior, often *subjectively* by humans or LLMs [124, 157, 262]. For specific scenarios, metrics like partisan bias and human likeness index are proposed [43]. When simulations target real-world scenarios with available empirical data, automated metrics like emotion, attitude, behavior consistency, and user taste alignment can be designed for more *objective* evaluations by comparing simulation content with real-world data [56, 148, 270].

Macro-level Evaluation. Social interactions often lead to collective outcomes, so it is important to evaluate whether macro-level outcomes show patterns and trends that are consistent with the real world. For sociology and online platforms, attention is typically given to whether the scale of propagation, the distribution and trends of collective opinions and traits align with those of the real world. In addition to qualitative methods such as *subjective* evaluation [56, 270], some studies have proposed quantitative metrics, such as fitted parameters, correlation coefficients and change of toxicity of community content to measure these differences *objectively* [137, 148, 202, 256]. Similarly, in economic simulation, the evaluation of simulated economic systems depends on whether they can reproduce the most representative macroeconomic laws [116].

System-level Evaluation. System-level evaluation is concerned with assessing the overall performance of a simulation system, irrespective of the specific content being simulated. With the growing number of agents in simulation, the focus of contemporary research has been on system efficiency and associated costs. Efficiency is assessed through various metrics, such as the time it takes to run a simulation, the resources that are utilized

Table 6. Comparative analysis of society simulation works: alignment, scalability, generalization, interpretability, and typical failure situations.

Architecture	Agent Engine	Representative Papers	Alignment	Scalability	Generalization	Interpretability	Failure
ABM	Rule-based	Schelling [180], HK [80], Axelord [8]	Weak	Strong	Weak	Strong	Behavioral Realism
	Learning-based	Q-learning ABM [20], Epstein et al. [50], MARL [284]	Moderate	Strong	Moderate	Moderate	Semantic Understanding
LLM+ABM	Hierarchical Hybrid	HiSim [148], AgentTorch [39], Li et al. [119]	Moderate	Strong	Moderate	Moderate	Boundary Artifacts
	Functional Hybrid	LELMA [142], Netlogo-LLM [145], AgentSociety [159]	Moderate	Strong	Weak	Moderate	Integration Coherence
LLM	Naive Prompting	Argyle et al. [6], Sun et al. [194], Economics Arena [69]	Weak	Moderate	Strong	Weak	Stereotype Bias
	Social/Psychology Prompting	Chuang et al. [42], S ² [56], Agent-trust [240]	Moderate	Moderate	Moderate	Moderate	Theory Validation
	Task-specific Prompting	Park et al. [157], OASIS [256], SocioVerse [281]	Strong	Weak	Weak	Weak	Domain Transferability

during the process, and how well the simulation can scale with an increasing number of agents [154, 210, 256]. These metrics are crucial for understanding how well the system can handle complexity and the demands of larger simulations. On the cost side, evaluations often center on the number of tokens consumed during the simulation or the financial expenditure incurred [262].

5.4 Discussion

Comparison of Simulation Schemas. Table 6 compares society simulation approaches across two key dimensions: architectural paradigm and agent reasoning engine. Traditional ABM offers strong scalability and transparency via rule-based design, but limited behavioral realism. Hybrid LLM+ABM approaches [39, 148, 159] enhance behavioral richness through modularized reasoning, at the cost of integration challenges such as boundary artifacts and cross-module inconsistency. Pure LLM-based simulations span diverse prompting strategies. Naive prompting enables broad generalization but risks amplifying social stereotypes [6, 194]. Theory-informed prompting induces more structured behavior, yet often relies on empirically unvalidated assumptions [42, 56]. Task-specific prompting achieves strong domain alignment but exhibits limited transferability [157, 256, 281]. Overall, for large-scale or long-horizon simulations, traditional ABM and hybrid architectures remain more robust to issues such as temporal behavioral drift [236] and limited interpretability [104, 269]. LLM-centric approaches are better suited for short-term, behaviorally rich simulations when carefully calibrated. Across paradigms, researchers must account for recurring failure modes, including stereotype amplification, validation gaps, and architectural fragility.

Reflection on Evaluation. Evaluation remains a core challenge in social simulation. At the micro level, most studies focus on coarse alignment between agent- and human-generated content [56, 279, 281]. Yet label-limited classification misses semantic consistency, whereas content overlap remains overly restrictive, necessitating finer-grained, semantically informed metrics. At the macro level, evaluation is even harder: scarce real-world data hinder direct comparison between collective outcomes of agents and actual groups. Existing attempts within specific scenarios [159, 256, 268] offer useful insights, but their generalizability remains unclear. In particular, assessing the emergence of collective phenomena is especially challenging: while some studies observe specific emergent patterns in controlled settings, it remains difficult to compare effectiveness across methods. Addressing this gap may require drawing on agent-based modeling traditions to design classical scenarios that support systematic evaluation and cross-paradigm comparison.

6 Datasets, Benchmarks and Tools

6.1 Individual Simulation

Table 7 summarizes commonly used datasets for individual simulation. Description datasets include individual-specific information, such as life experiences, relationships, and demographic information, typically sourced from literature or Wikipedia. Dialogue datasets contain conversations from specific scenarios, gathered from various sources.

Table 7. Summary of commonly used datasets for individual simulation.

Domain	Dataset	Type	Source	# individual num	# dialogue num	Paper	Link
Characters	LISCU	Description	Books, Summaries	9,499	/	[18]	Link
	FoCus	Description	Wikipedia	/	86,712	[89]	Link
	HPD	Dialogue/Description	Books	1	about 2,500	[26]	Link
	LaMP	Description	/	/	/	[179]	Link
	Multimodal Persona Chat	Image/Dialogue	Reddit	/	15,000	[4]	Link
	LiveChat	Description/Dialogue	Douyin	351	1,330,000	[59]	Link
	COMSET	Dialogue	Strips	13	53,903	[2]	Link
	ChatHaruhi	Dialogue	Movies, Script	32	54,000	[108]	Link
	RoleBench	Dialogue	Scripts	100	168,093	[224]	Link
	CharacterLLM	Description	/	9	14,400	[183]	Link
	CharacterDial	Description/Dialogue	Literary Resources, LLM, Human	250	1,034	[289]	Link
	Synthetic Persona Chat	Description/Dialogue	LLM	10,371	21,907	[88]	Link
	RoleEval	Description	Wikipedia, Baidu, Fandom, Moegirlpedia	300	6,000	[185]	Link
	CharacterEval	Description/Dialogue	Novels, Scripts	77	1,785	[203]	Link
	Cross	Description	Books	/	/	[265]	Link
Demographics	MMRole	Description/Dialogue/Image	Wikipedia, Baidu	85	14,000	[44]	Link
	RP	Dialogue	Novels, Scripts	331	3,552	[263]	Link
	MPI	Description	/	/	/	[95]	Link
	Who is GPT3 Dataset	/	/	/	/	[144]	Link
	EmotionBench	/	/	/	/	[86]	Link
	OpinionQA	/	Surveys	/	/	[113]	Link
	PersonaHub	Description	LLM	200,000	375,000	[61]	Link

6.2 Scenario Simulation

Table 8 summarizes commonly used datasets for scenario simulation. The datasets include various formats such as QA, multiple-choice, code, and game. We can observe that QA and multiple-choice formats dominate the data types, while domain-specific datasets like judicial, game, and media prefer to preserve domain-tailored data types.

6.3 Society Simulation

Table 9 summarizes commonly used datasets for society simulations. They serve for various simulation objectives, sourced from various channels, such as public surveys, existing datasets and online platforms. They typically contain profiles to initialize agents and some evaluation data to provide real-world behavior references.

6.4 Commonly Used Agent Frameworks

Table 10 summarizes commonly used tools for building LLM-based agents. *Frameworks* provide base classes for agents and environments, requiring users to implement only task-specific functions. *Platforms* build on frameworks by adding interfaces and infrastructure for interaction. *Products* are domain-specific or consumer-facing applications. These tools vary in deployment scale: some target *small* settings (fewer than a dozen agents) with fine-grained modeling. Others support *large* simulations involving thousands to millions of agents for studying emergent behaviors. A third group, labeled *any*, allows user-defined scaling but often lacks special optimizations for efficient large-scale execution.

7 Potential Risks

While LLM-based agents enable rich simulations across individual, scenario, and society levels, they also introduce a spectrum of potential risks. They arise naturally from the layered structure of the simulations and can propagate from individual behaviors to collective and societal dynamics. In this section, we discuss key risks and strategies to mitigate them, organized to highlight their manifestation at different simulation scales.

Reproducibility. Reproducibility is a key challenge in LLM-driven simulations across all levels. At the individual level, outputs are sensitive to random seeds, prompt phrasing, and minor model variations [62, 296]. In scenario simulations, interactions among heterogeneous agents amplify non-determinism, as small local deviations can cascade. At the societal level, slight changes in initialization, network structure, or population composition may

Table 8. Summary of commonly used datasets for scenario simulation.

Domain	Datasets	Type	Complexity	# case	Collection	Used by	Data Link
Dialog-Driven	MiniWob++	Web Interaction	Hard	/	human	[231]	Link
	SOTOPIA	Open-Ended Environment	Hard	/	human	[291]	Link
	WebQSP	QA	Easy	4,737	human	[139]	Link
	FairEval	QA	Medium	80	human	[23]	Link
	MMLU	Multiple-Choice	Hard	115,700	real world	[49, 198, 200, 267, 275]	Link
	BIG-bench	/	Hard	/	human	[31, 49, 275]	Link
	MetaQA	QA	Medium	407,513	real world, human	[139]	Link
	MATH	Math Problem	Medium	12,500	real world	[231]	Link
	Jubensha	Game	Medium	1,115	real world	[230]	Link
	ALFWorld	Embodied Environment	Medium	3,827	human	[231]	Link
	Topical-Chat	Conversational	Medium	10,784	human	[23]	Link
	CSQA	Multiple-Choice	Easy	1,221	human	[245]	Link
	PIQA	Multiple-Choice	Easy	1,838	human	[198, 245]	Link
Task-Driven	StrategyQA	Multiple-Choice	Easy	2,290	human	[245]	Link
	HumanEval	Code	Easy	164	real world	[31, 47, 82, 267]	Link
	MBPP	Code	Easy	974	real world	[47, 82]	Link
	SRDD	Software Requirement	Medium	1,200	synthetic	[162–164, 267]	Link
	PubMedQA	QA	Medium	273,500	human, synthetic	[200]	Link
	MedQA	Multiple-Choice	Medium	61,097	real world	[151, 200]	Link
	MedMCQA	Multiple-Choice	Hard	194,000	real world	[200]	Link
	MVME	Medical Record	Medium	506	real world	[51]	Link
	GSM8K	Math Problem	Easy	8,500	human	[49, 266]	Link
	MGSM	Math Problem	Hard	2,750	human	[31]	Link
	Math	QA	Hard	50,000	synthetic	[110, 275]	Link
	SimuCourt	Legal Cases	Medium	420	real world	[78]	Link
	KINLED	Conversational	Medium	10,546	human, synthetic	[192]	Link
	CommonGen-Hard	Concept	Easy	200	human	[31, 267]	Link
	HMDB51	Video Clip	Medium	13,320	real world	[244]	Link

yield widely different emergent behaviors [156]. Mitigating these issues requires rigorous practices, including standardized reporting, transparent experimental documentation, and statistical frameworks that account for inherent variability rather than relying solely on deterministic replication.

Bias. While some biases, such as cultural or demographic traits, may enhance realism, harmful biases can distort outcomes, reduce fairness, and compromise validity. At the individual level, they arise from training data [71, 128], prompts, or user interactions [252], manifesting as reasoning shortcuts, political imbalances [76, 177], or stereotyped roles. In scenario simulations, these biases can accumulate through agent interactions, reinforcing flawed group decisions and propagating role-based stereotypes. At the societal level, systemic biases in initial conditions or network structures may amplify polarization or create echo chambers. Mitigation strategies include systematic auditing, targeted debiasing during training, transparent evaluation, and deliberate incorporation of heterogeneity to prevent over-amplification.

Interpretability. The black-box nature of foundation models makes LLM-based simulations difficult to interpret [237]. Although mechanistic interpretability reveals internal circuits and causal structures [126, 170], it offers limited insight into agent behavior within simulations. At the individual level, agents may reproduce roles or traits without clear evidence of underlying reasoning. Techniques such as chain-of-thought reasoning [225] and self-explanation [208] partially improve action transparency. At the scenario and societal levels, tracing multi-agent dynamics and emergent outcomes remains challenging, motivating approaches such as role-based

Table 9. Summary of commonly used datasets for society simulation. Init. means the data provides profile to initialize agents, and Eval. means it provides data to validate the simulation effectiveness.

Scenario	Dataset	Init.	Eval.	Content	# case	Simulation Objectives	Used by	Data Link
General Economics	2018 U.S. population	✓		profile	100 people	macroeconomic activities	[116]	Link
	public government data		✓	rent information	51 users	resource allocation	[92]	Link
	big-five-data	✓		profile	1,000 people	epidemic modeling	[228]	Link
	American Community Survey	✓		profile	8.4M people	epidemic modeling	[39]	Link
	Bureau of Labor Statistics		✓	labor statistics	8.4M people	unemployment rate	[39]	Link
Sociology and Political Science	ANES	✓	✓	profile,answer	15,626 responses	voting	[6, 195, 279]	Link
	Pigeonholing Partisans	✓	✓	profile,answer	2,107 responses	partisan bias	[6]	Link
	Global Warming	✓	✓	profile,answer	2,310 responses	opinion	[?]	/
	Interview	✓	✓	profile,answer	1,002 users	opinion and behavior	[158]	Link
	Ultimatum Game		✓	money allocation	10,000 pairs	money allocation	[3]	Link
	Wisdom of Crowds		✓	answers to questions	15,000 answers	wisdom of crowds	[3]	Link
	Milgram Shock Experiment		✓	behavior records	100 people	obedience behavior	[3]	Link
	15 Topics	✓		profile, opinion	10 users	opinion dynamics	[42]	Link
	Becker-2019	✓	✓	profile, answers	1,120 users	wisdom of crowds	[43]	Link
	Controversial Beliefs Survey	✓	✓	profile, opinion	564 users	opinion	[41]	/
Online Platforms	Echo Chambers	✓		profile	3 networks	opinion polarization	[206]	/
	Gender Discrimination	✓	✓	profile, opinion	8,563 users	opinion dynamics	[56]	/
	ANES	✓		profile	500 users	partisan bias	[202]	Link
	SAGraph	✓	✓	profile, interaction	40300 influencers	influencer selection	[280]	/
	Metoo	✓	✓	profile, opinion	1,000 users	opinion dynamics	[148]	Link
	Roe	✓	✓	profile, opinion	1,000 users	opinion dynamics	[148]	Link
	BLM	✓	✓	profile, opinion	1,000 users	opinion dynamics	[148]	Link
	Twitter15	✓	✓	profile, behavior	198 news	rumor propagation	[256]	Link
	Reddit	✓	✓	profile, comment	116,932 comments	herd effect	[256]	/
	Steam	✓	✓	profile,interaction	281,205 users	interaction	[87, 270]	Link
	MovieLens	✓	✓	profile,interaction	298,074 users	interaction	[87, 210, 270]	Link
	Amazon Review CD	✓	✓	profile,interaction	100 users	interaction	[274]	Link

Table 10. A list of representative frameworks and applications to build your agent. We categorize the agent frameworks according to their type, agent scale, modality, and tool-learning.

Domain	Type	Name	Agent Scale	Multi-Modal	Tools	Paper	Link
Individual	Product	AutoGPT	Small			[255]	Link
	Product	Character.ai	Small	✓		/	Link
	Product	BabyAGI	Small			[27]	Link
	Framework	HuggingGPT	Small	✓		[186]	Link
Scenario	Framework	AutoGen	Any	✓	✓	[231]	Link
	Framework	AgentVerse	Any		✓	[31]	Link
	Product	AI Dungeon	Small			/	Link
	Framework	metaGPT	Small			[82]	Link
Society	Platform	AgentTorch	Large			[39]	Link
	Framework	SocioVerse	Large			[281]	Link
	Framework	YuLan-OneSim	Large			[213]	Link
	Framework	OASIS	Large		✓	[256]	Link
	Framework	AgentSociety	Large	✓	✓	[159]	Link
General	Framework	LangChain	Any	✓	✓	/	Link
	Platform	AGiXT	Any	✓	✓	/	Link
	Framework	CAMEL	Any	✓	✓	[110]	Link
	Framework	AgentScope	Any	✓	✓	[58]	Link
	Framework	Pydantic AI	Any	✓	✓	/	Link
	Platform	LazyLLM	Any	✓	✓	/	Link

attribution, interaction visualization, and structured debate. Overall, systematic mechanisms for reasoning transparency and decision traceability across modeling scales are essential for producing credible and actionable simulation results.

Safety. Safety in LLM-driven simulations focuses on preventing harmful, misleading, or inappropriate outputs, with toxicity [204] and hallucination [93] as key risks. At the individual level, unsafe outputs may arise from hallucinated details or persona-driven toxicity [29, 73], especially when stereotypes or controversial roles are involved. In scenario simulations, harmful content can propagate through agent interactions, polluting shared memory and disrupting collective reasoning [67, 138]. At the societal level, unsafe signals may escalate into systemic distortions, fueling polarization or large-scale misinformation. Mitigation requires multi-layered defenses, including toxicity and hallucination detection, inter-agent consistency checks, and systemic auditing, to keep simulations credible and aligned with intended purposes.

Misuse Risk. Misuse risk refers to adversarial exploitation of LLM-based simulations. At the individual level, prompt injection [65] and jailbreaking [297] can bypass safety mechanisms, exploiting persona-driven role-playing to produce unsafe or deceptive outputs. In scenario simulations, adversaries may coordinate multiple agents to automate harmful tasks [49, 197], such as unsafe software and product design. At the societal level, misuse can escalate into systemic threats [237], including large-scale psychological manipulation. Mitigation requires both technical safeguards, such as adversarial robustness checks, persona filtering, cross-agent verification, and governance measures, such as regulatory oversight, accountability frameworks, and transparent reporting.

8 Challenges and Future Directions

8.1 Individual Simulation

Despite significant progress in individual simulation using LLMs, current approaches still face several key challenges. (1) Many methods attempt to replicate surface-level behaviors or attributes, such as demographic traits [153, 178], but **the lack of complete offline behavioral histories and internal thoughts for target individuals makes high-fidelity replication difficult**, often resulting in the reproduction of biases, stereotypes, and hallucinations rather than deeper cognitive patterns. (2) Since LLMs are trained on general corpora, they tend to exhibit an “average personality,” which **limits the behavioral heterogeneity** of simulated populations [236]. (3) Furthermore, in dynamic, context-specific simulations covering dialogues, games, and embodied tasks [85, 127, 166, 208, 233], LLMs struggle to maintain **coherent decision-making over time**, adapt to continuously changing environments, and generalize beyond predefined rules. (4) Finally, current approaches face **mass-production challenges**, as it remains difficult to construct large numbers of fine-grained individual simulators while preserving alignment and behavioral fidelity.

8.2 Scenario Simulation

Despite steady progress in scenario-level simulation using LLM-based agents, several critical challenges remain. (1) Current approaches often require **human expertise to design flows and structures**, including decomposing tasks into multiple stages or defining domain-specific agents [151, 162, 197]. While these human-designed frameworks improve controllability and feasibility, they hinder the ability of systems to **automatically adapt to new or unseen scenarios**, limiting generalization and scalability. (2) In collaborative settings, LLM-based agents often exhibit a **high failure rate in coordination**. Recent research [22] further categorizes these failures into 14 failure modes spanning specification, inter-agent misalignment, and task verification, highlighting the complexity of building reliable multi-agent systems. (3) Finally, scenario simulation also faces a persistent challenge in balancing **accuracy and efficiency**. High-fidelity simulations such as those involving medical workflows [51] or complex social interactions [253] often demand significant computational and design resources, while efficiency-oriented methods introduce simplifications that compromise realism [122, 273]. Determining how to trade off between these competing objectives, and whether self-adaptive mechanisms can mitigate such trade-offs, remains an open research problem.

8.3 Society Simulation

Despite considerable progress in society simulation using LLM-driven agents, current approaches face several key challenges. (1) **Alignment with real world remains limited.** Many studies simply observe agent behaviors within constructed environments without ensuring alignment with real and dynamic social contexts, including realistic population composition and interaction mechanisms [281]. As a result, it is unclear whether their conclusions have practical significance. Although some work tries to address alignment in specific social scenarios, such as social networks and economic models [56, 116], these efforts are constrained by data sparsity, incompleteness, and noise, making it difficult to verify whether agent populations truly replicate human social dynamics. (2) **Scaling up** to large populations and more complex interactions currently requires partial simplifications of the simulation [39, 68, 148]. While this trade-off is necessary for feasibility, determining how to minimize the sacrifice of simulation fidelity remains an open challenge. (3) **LLM interpretability** poses another difficulty: the black-box nature of LLMs makes it hard to provide rigorous causal explanations for individual behaviors or collective outcomes, and developing more transparent and robust approaches at the intersection of LLMs and social science remains a critical research problem.

8.4 Trade-offs and mutual promotion between different levels

When advancing from individual to scenario and society simulations, researchers face inherent trade-offs between **fidelity** and **scalability**. Thus, micro-level studies prioritize fidelity, while large-scale evaluations emphasize scalability, with both requiring basic practical feasibility, such as computational efficiency, data availability, and calibration reliability, before optimizing another dimension. Recent work on lightweight personalized preference modeling [17, 115] has proposed adapter-based approaches that enhance scalability while preserving fidelity. In contrast, for large-scale societal simulations, strict individual alignment, though beneficial, is not always critical for obtaining valid macroscopic outcomes. Large-scale experiments further suggest that moderate individual-level noise can be tolerated as long as aggregate patterns remain reliable, although interpretations should be carefully bounded to avoid overextending conclusions beyond the simulation's scope [236].

Beyond trade-offs, the three levels can also **mutually reinforce each other**. Individual simulation forms the foundation: enhancing a model's general task-solving and role-playing abilities often improves the effectiveness of scenario and large-scale societal simulations [61, 199]. In turn, scenario simulations provide diverse and challenging environments, where interactive feedback fosters individual learning [13, 215] and improve the individual model's ability to achieve goals in the scenarios. Finally, societal simulations reveal behavioral patterns and crowd wisdom, which can further guide profile rewriting at the individual level and coordination strategies in scenario-based group interactions.

9 Conclusion

In this paper, we categorize LLM-driven social simulations into three types: individual, scenario, and society simulation, highlighting their progression from modeling individual behaviors to replicating complex social dynamics. By systematically reviewing architectures, methods, and evaluations across these categories, we provide a structured framework for advancing research in this field. This work aims to guide the development of LLM-based simulations and foster interdisciplinary studies to address real-world challenges and support decision-making.

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